

Base Rate Neglect Clinic Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	base-rate-neglect-clinic

Big idea: Probability judgements depend on the background rate as well as the new evidence.

Overview

Students diagnose why a vivid test result can mislead when the underlying base rate is ignored.

Student Prompt

Inspect Base Rate Neglect Clinic Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Base Rate Neglect Clinic Stimulus A	Student-facing stimulus 1: In 1000 people, 10 have Condition X. The test catches 9 of them but falsely flags 99 people without it. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Base Rate Neglect Clinic Stimulus B	Student-facing stimulus 2: Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Base Rate Neglect Clinic Reveal / Twist	Student-facing stimulus 3: Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Probability judgements depend on the background rate as well as the new evidence.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Base Rate Neglect Clinic Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Base Rate Neglect Clinic Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does representation affect mathematical knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how probability judgements depend on the background rate as well as the new evidence.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use natural frequencies rather than formulas at first.
- Keep the scenario fictional and low-stakes while preserving the structure of diagnostic reasoning.

Ready-to-Use Examples

- In 1000 people, 10 have Condition X. The test catches 9 of them but falsely flags 99 people without it.
- Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?

Suggested Timing

Stage	Teacher move	Watch for
1	Present a fictional condition with a low base rate and a fairly accurate test.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Ask students to estimate the chance someone has the condition after a positive result.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Build the scenario with a 1000-person or 100-person frequency table.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Compare intuitive estimates with the frequency result.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why base rates are easy to neglect.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- The test sounds accurate, but we need to ask: accurate among how many real cases?
- The denominator is doing quiet but essential knowledge work.

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Support: use 100 people and round numbers.
- Challenge: ask students to calculate a second scenario with a higher base rate.

Extension Tasks

- Students rewrite a risk claim in three formats and evaluate which is most responsible.
- Compare this with legal evidence, medical testing, and spam filtering.

Related Activities

- Monty Hall Shock
- Calculator Trust Test
- P-hacking with Random Data